Experiment4a:

CODE:

```
  GNU nano 8.7.1
#!/bin/bash
echo "Currently working users are"
echo ""
who
echo ""

read -p "Enter username: " user

login_time=$(who | awk -v u="$user" '$1 == u {print $3, $4; exit}')

if [ -z "$login_time" ]
then
    echo "User is not currently logged in."
else
    echo "Login time: $login_time"

    login_seconds=$(date -d "$login_time" +%s)
    current_seconds=$(date +%s)

    total_seconds=$((current_seconds - login_seconds))

    hours=$((total_seconds / 3600))
    minutes=$(((total_seconds % 3600) / 60))

   echo "Total working time: $hours hours $minutes minutes"
fi
```

OUTPUT:

```
tofiq@Tofiq:~$ bash exp4a.sh
Currently working users are


Enter username: tofiq
User is not currently logged in.
```

Experiment4b:

CODE:

```
  GNU nano 8.7.1
#!/bin/bash

if [ "$#" -ne 3 ]
then
    echo "Usage: $0 filename start_line end_line"
fi

file="$1"
start="$2"
end="$3"

sed -n "${start},${end}p" "$file"
```

OUTPUT:

```
tofiq@Tofiq:~$ bash exp4b.sh numbers.txt 2 7
2
3
4
5
6
7
```

Experiment4c:

CODE:

```
  GNU nano 8.7.1
#!/bin/bash

if [ "$#" -lt 2 ]
then
    echo "Need atleast two arguments"
fi

word="$1"
shift

for file in "$@"
do
    if [ -f "$file" ]
    then
        sed -i "/$word/d" "$file"
        echo "Deleted lines containing '$word' from $file"
    else
        echo "$file does not exist."
    fi
done
```

OUTPUT:

```
tofiq@Tofiq:~$ bash exp4c.sh Sharma NAME1.TXT
Deleted lines containing 'Sharma' from NAME1.TXT
tofiq@Tofiq:~$ cat NAME1.TXT
Virat Kohli
Jasprit Bumrah
Mohd Siraj
Mohd Shami
```